

Exploratory Data Analysis (EDA)

1. What is the average number of orders per customer? Are there high-value repeat customers?

The dataset contains 830 orders and 91 customers, resulting in an average of approximately 9.12 orders per customer. Analysis of customer purchase behavior indicates the presence of several high-value repeat customers. Customers such as QUICK-Stop, Ernst Handel, and Save-a-lot Markets generated significantly higher sales revenues while placing multiple orders, making them key contributors to overall business revenue.

2. How do customer order patterns vary by city or country?

Customer order patterns vary considerably across geographic locations. The United States generated the highest sales revenue, followed by Germany and Austria. Major cities contributed a larger share of orders and revenue compared to smaller cities. This suggests that customer demand is concentrated in a few high-performing markets, highlighting opportunities for targeted marketing and customer retention strategies.

3. Can we cluster customers based on total spend, order count, and preferred categories?

Customers can be grouped into three major segments:

1. High-Value Customers – High spending and frequent purchases.
2. Regular Customers – Moderate spending and moderate order frequency.
3. Low-Value Customers – Infrequent purchases and lower spending.

The high-value segment contributes a disproportionately large share of total revenue and should be prioritized for loyalty programs and retention initiatives.

4. Which product categories or products contribute most to order revenue?

Beverages emerged as the highest revenue-generating category, followed by Dairy Products and Confections. Several premium products consistently generated higher revenues due to strong demand and pricing advantages. Revenue concentration among a few categories indicates the importance of maintaining inventory and supplier relationships for these products.

5. Are there any correlations between orders and customer location or product category?

A positive relationship exists between customer location and order volume. Countries with larger customer bases generated more orders and higher revenue. Product category also influences purchasing behavior, with categories such as Beverages and Dairy Products attracting higher demand and contributing significantly to sales performance.

6. How frequently do different customer segments place orders?

High-value customers place orders more frequently and maintain long-term purchasing relationships with the company. Regular customers contribute consistent business activity, while low-value customers place orders less frequently. This pattern suggests that customer segmentation can be used to design targeted engagement and retention strategies.

7. What is the geographic and title-wise distribution of employees?

Employees are distributed across multiple regions, with a significant concentration in the United States. Sales Representatives represent the largest employee group, followed by managerial and executive positions. This structure reflects the organization's emphasis on sales and customer relationship management.

8. What trends can we observe in hire dates across employee titles?

Employee hiring occurred over multiple years, indicating gradual organizational growth. Senior roles generally correspond to earlier hire dates, while newer hires are more common among operational and sales positions. This trend reflects a stable organizational hierarchy and workforce expansion over time.

9. What patterns exist in employee title and courtesy title distributions?

Sales Representatives account for the largest proportion of employees. Courtesy titles such as Mr., Ms., and Dr. are distributed across different employee groups. The distribution indicates a diverse workforce with varying levels of experience and responsibility.

10. Are there correlations between product pricing, stock levels, and sales performance?

Products with competitive pricing and adequate stock levels generally achieved stronger sales performance. Higher-priced products generated substantial revenue but often experienced lower sales volumes. Maintaining optimal inventory levels is important to prevent stock shortages and maximize revenue opportunities.

11. How does product demand change over months or seasons?

Monthly sales analysis shows fluctuations throughout the year, with periods of increased demand followed by temporary declines. Sales generally trend upward over time, indicating business growth. Seasonal demand patterns suggest opportunities for inventory planning and promotional campaigns during peak periods.

12. Can we identify anomalies in product sales or revenue performance?

Several products exhibited unusually high sales volumes or revenue compared to category averages. These anomalies may result from promotional campaigns, seasonal demand spikes, or strong customer preferences. Identifying such products can help improve forecasting and inventory management.

13. Are there any regional trends in supplier distribution and pricing?

Suppliers are distributed across multiple countries, with a concentration in Europe and North America. Regional pricing differences were observed, likely due to sourcing costs, transportation expenses, and market conditions. These differences influence product profitability and procurement decisions.

14. How are suppliers distributed across different product categories?

Supplier participation varies across product categories. Categories such as Beverages and Condiments involve a larger number of suppliers, while specialized categories rely on fewer suppliers. Diversified supplier networks help reduce supply chain risk and improve product availability.

15. How do supplier pricing and categories relate across different regions?

Supplier pricing varies by region and category. Certain regions specialize in premium products with higher prices, while others provide cost-effective alternatives. Understanding these regional pricing patterns supports strategic sourcing and supplier selection decisions.